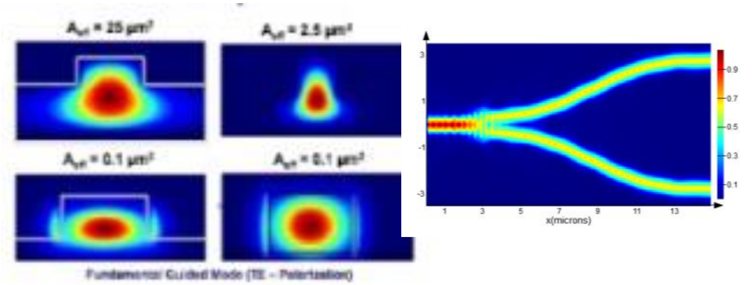
Eye Diagram Analysis

Outline

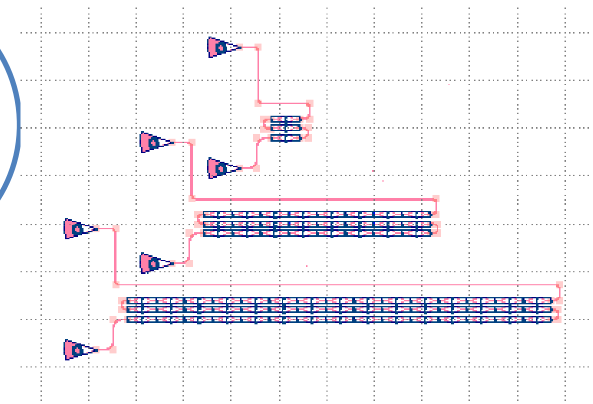
- **Eye Diagram Analysis**
  - **Introduction**
  - **Bit Error Rate (BER)**
  - **Pseudorandom Binary Sequence (PRBS)**
  - **Noise & Jitter**
  - **Lumerical Interconnect**
  - **Quality factor (Q- Factor)**

Steps for realizing Photonic Integrated Circuits (PICs)

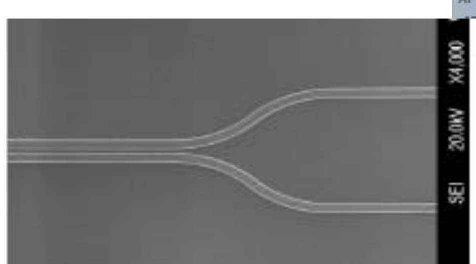
Modeling & Design

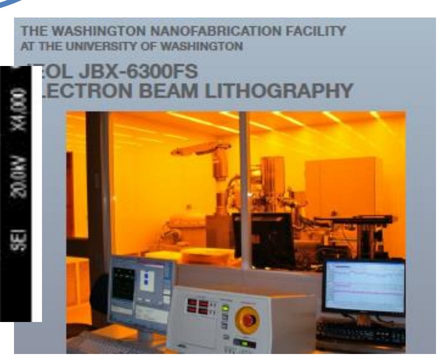


Mask Layout

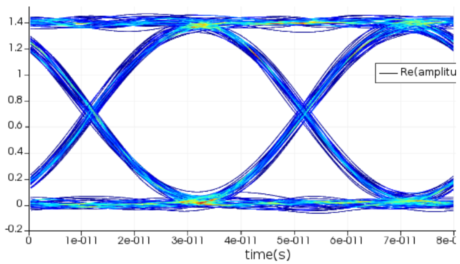


Fabrication





Test



[1]

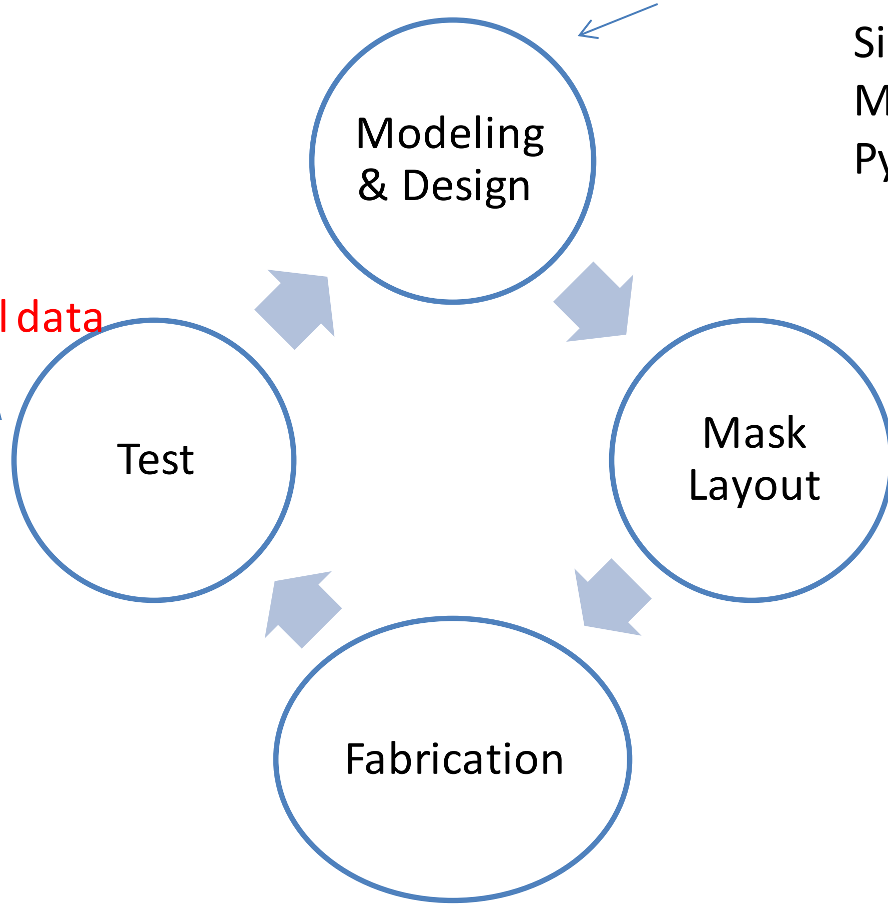
Charactering a chip performance based on modeling & experimental data (Eye diagram)

1- Based on Modeling data

Simulations tools
MATLAB
Python

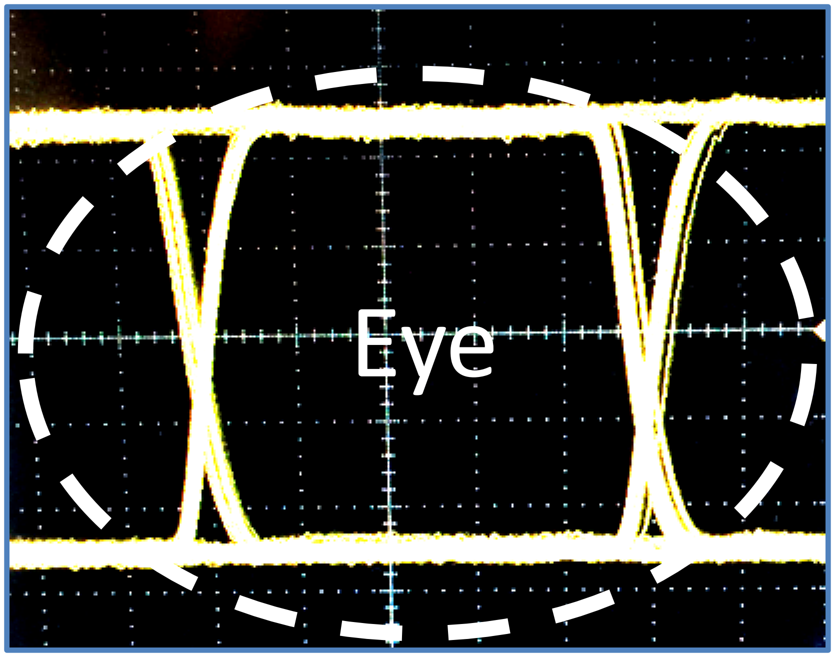
2- Based on Experimental data

Photonic probe station
MATLAB
Python



Eye diagram analysis: basic definition

Eye Diagram: an experimental method to measure and analyze an optical data at receiver side. For example to analyze the output data of an PIC devise.



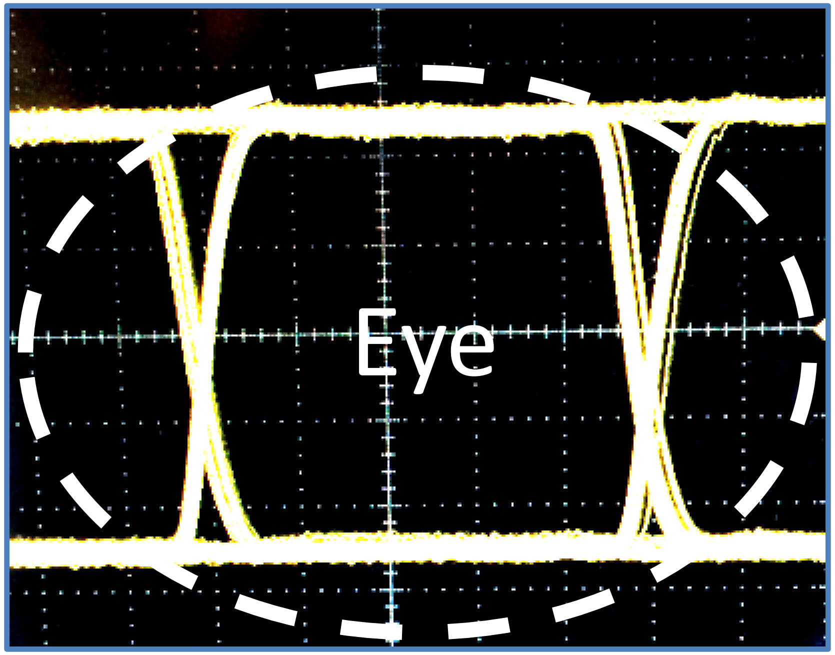
Eye diagram analysis: purpose

Eye Diagram: an experimental method to measure and analysis an optical data at receiver side. For example to analyze the output data of an PIC devise.

The main goal of an Eye Diagram analysis is to characterize the quality of data of an optical system

We can measure

1- Bit error Rate (BER)
2- Q factor



Bit error rate (BER): mathematical representation

$$\text{BER} = \frac{\text{Received error bits (n)}}{\text{Total number of received bits (N)}} \text{, Units: [-]}$$

n	N	BER
1	1E3	1E-3
1	1E6	1E-6
1	1E9	1E-9
1	1E12	1E-12
1	1E15	1E-15
1	1E24	1E-24

Bit error rate (BER): example

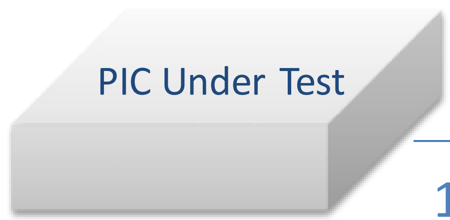
$$BER = \frac{\text{Received error bits } (n)}{\text{Total number of received bits } (N)}$$, Units: [-]

Example:

Sent bits

11001100110011001100

Total 20 sent bits



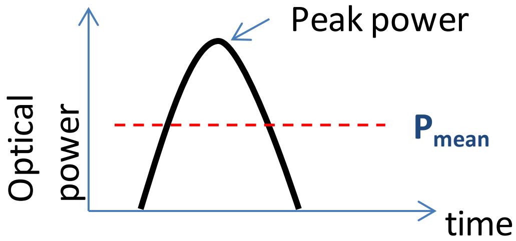
Received bits

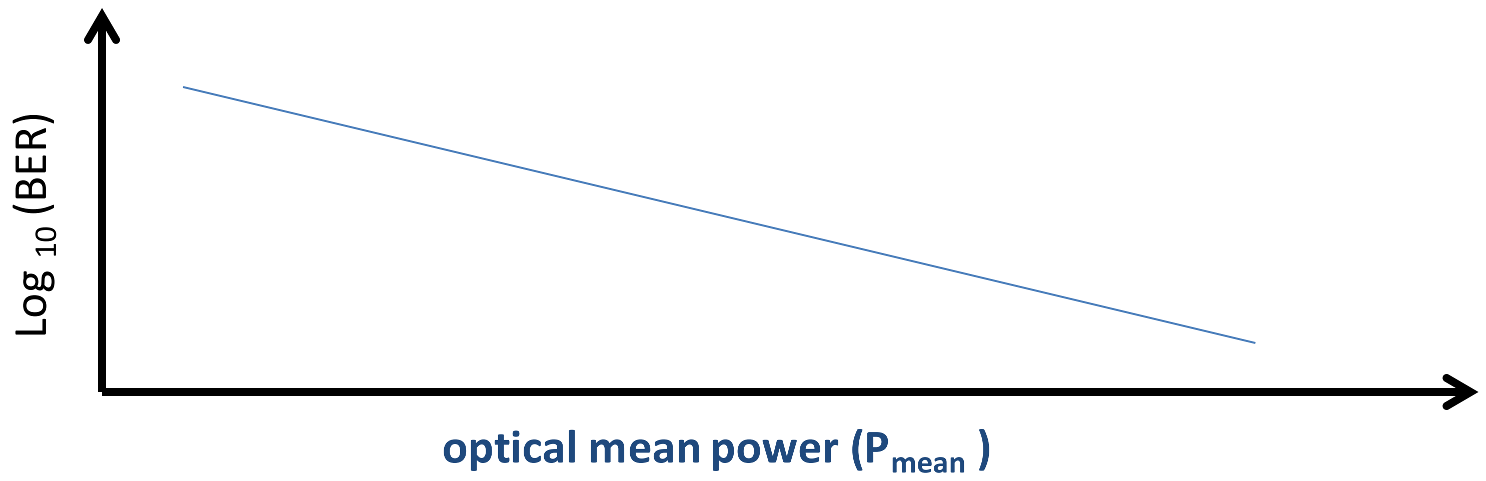
11001100010001001100

error

BER = 2/ 20= 0.1

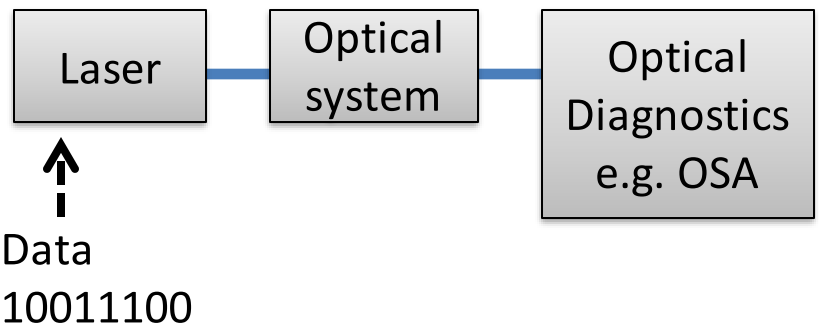
BER & optical mean power P_{mean}





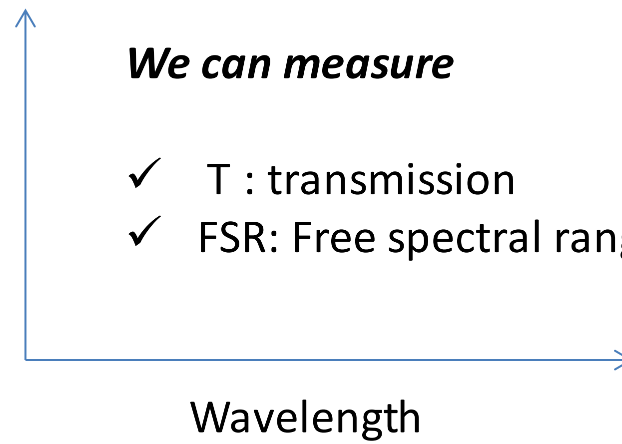
Spectral characterization approache

Optical Domain



We can measure

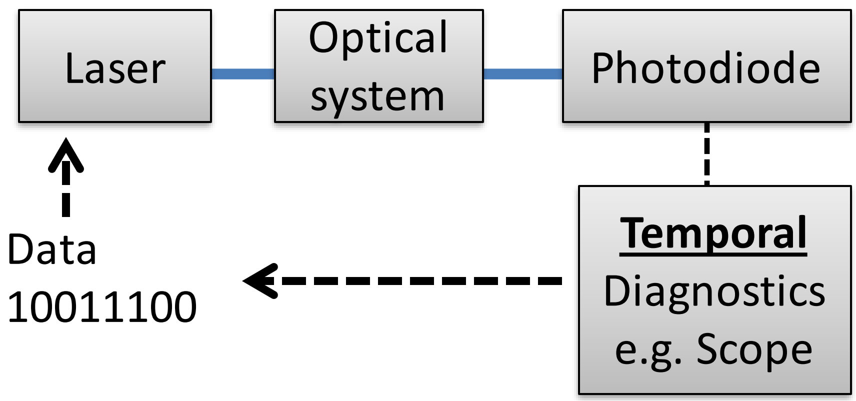
- ✓ T : transmission
- ✓ FSR: Free spectral range

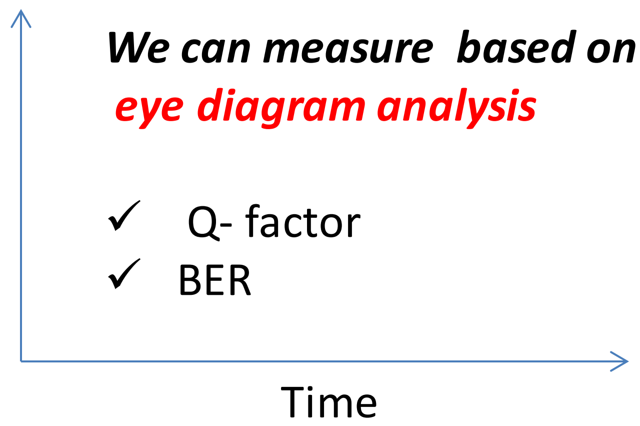


Key
Optical cable
Electrical cable

Temporal characterization approache

<u>Temporal Domain</u>



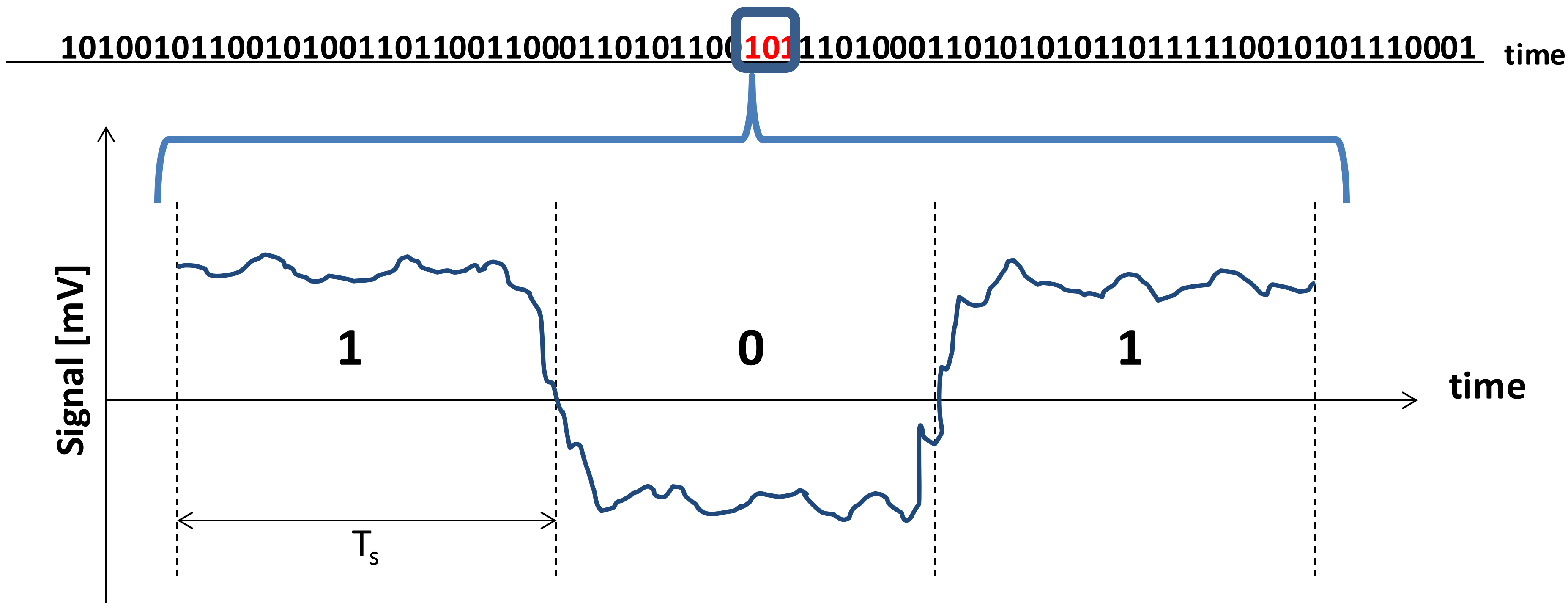


Key
Optical cable
Electrical cable

Eye Diagram Measures the Quality of Optical Received Signals

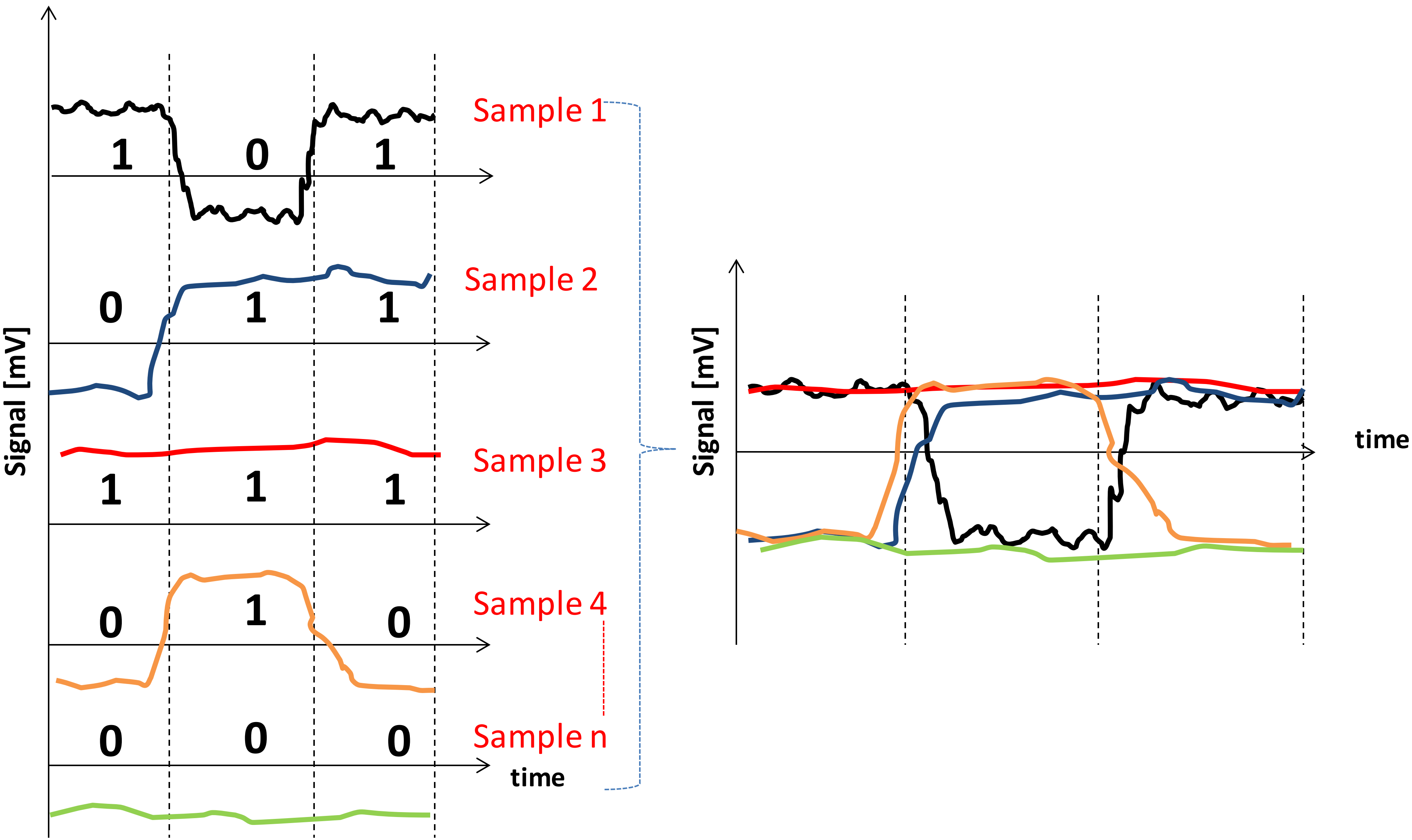
Sample of PRBS signals

Pseudorandom Binary Sequence (PRBS) data



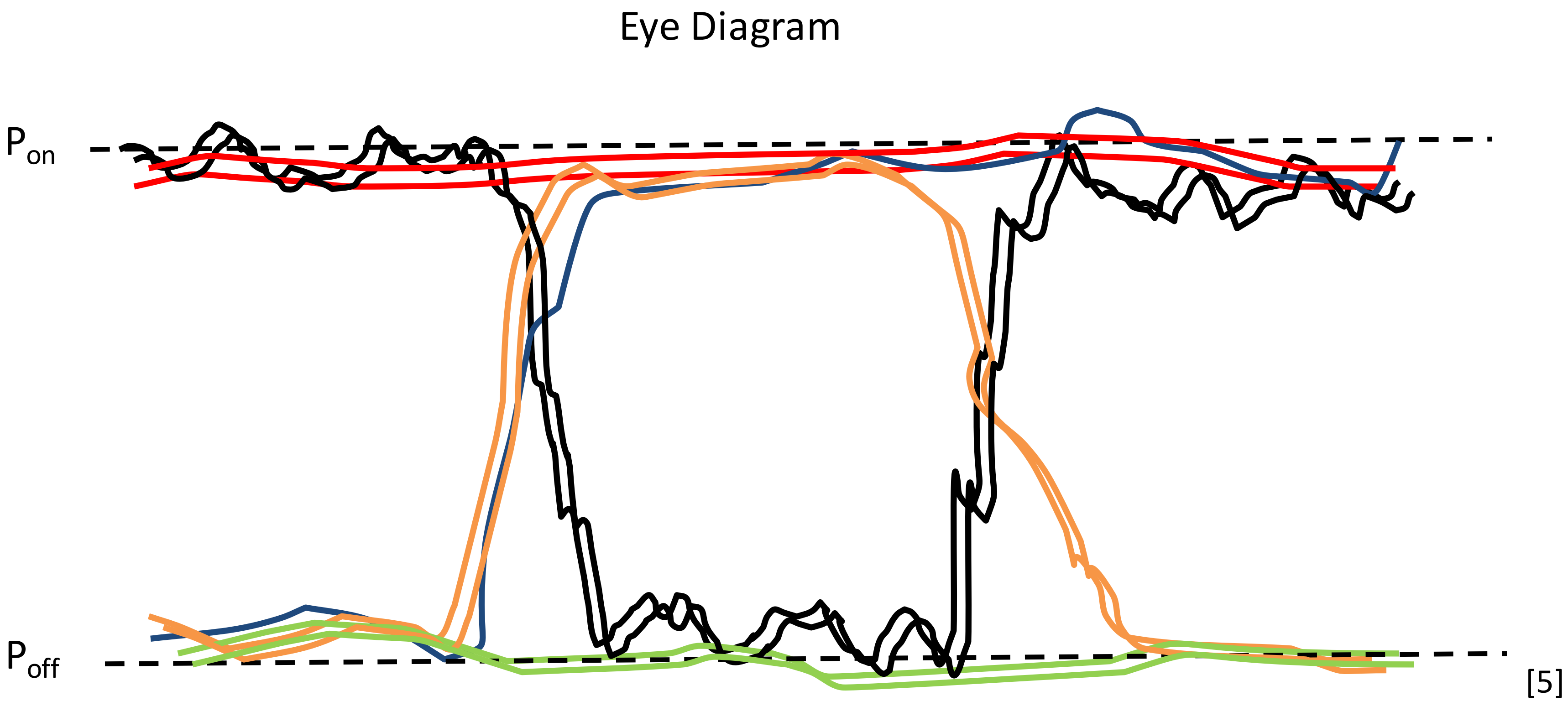
- Example: NRZ

Oscilloscope generates an eye diagram by composing several samples signals

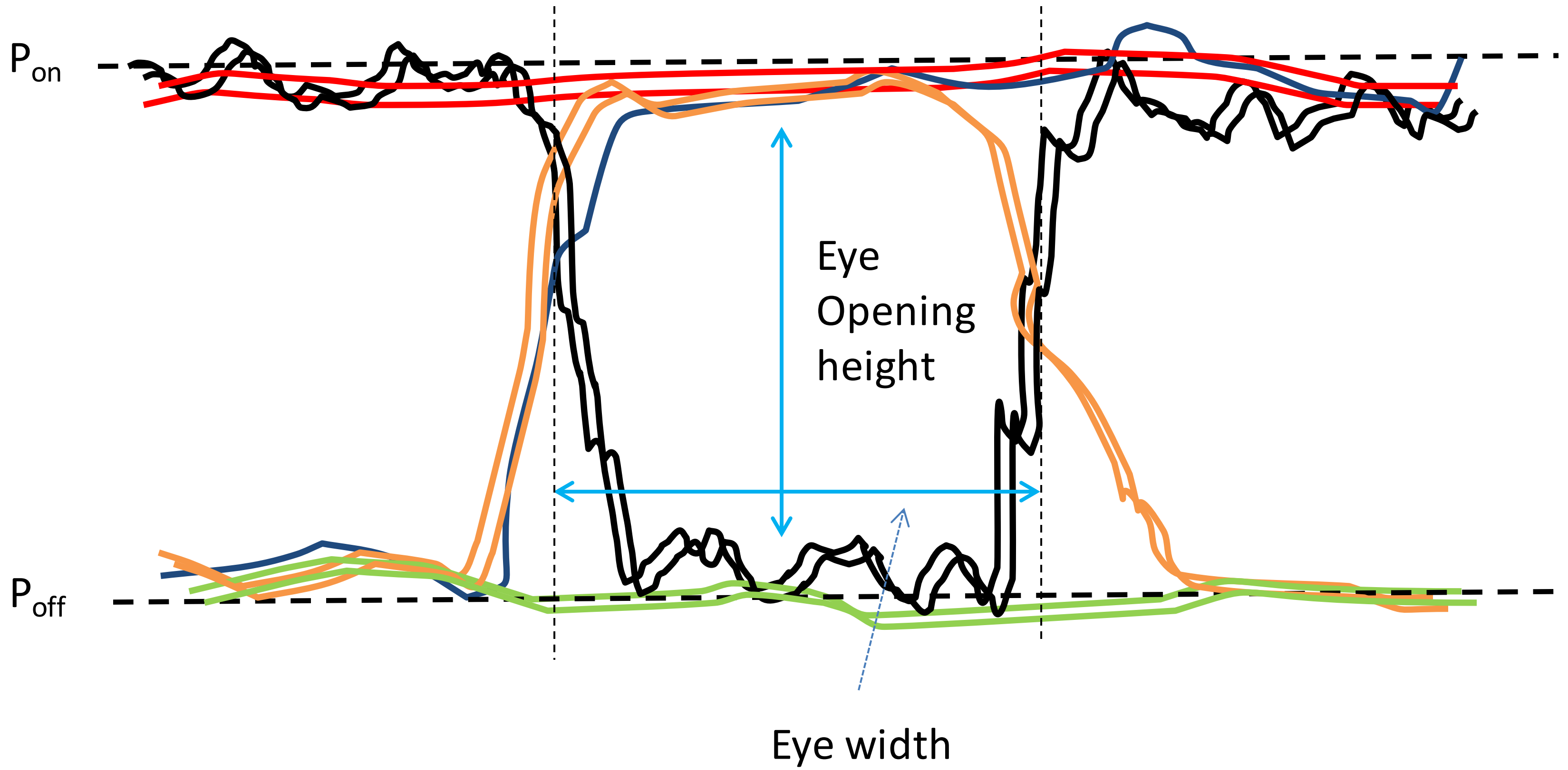


- Example: NRZ

Eye diagram analysis: on/ off levels

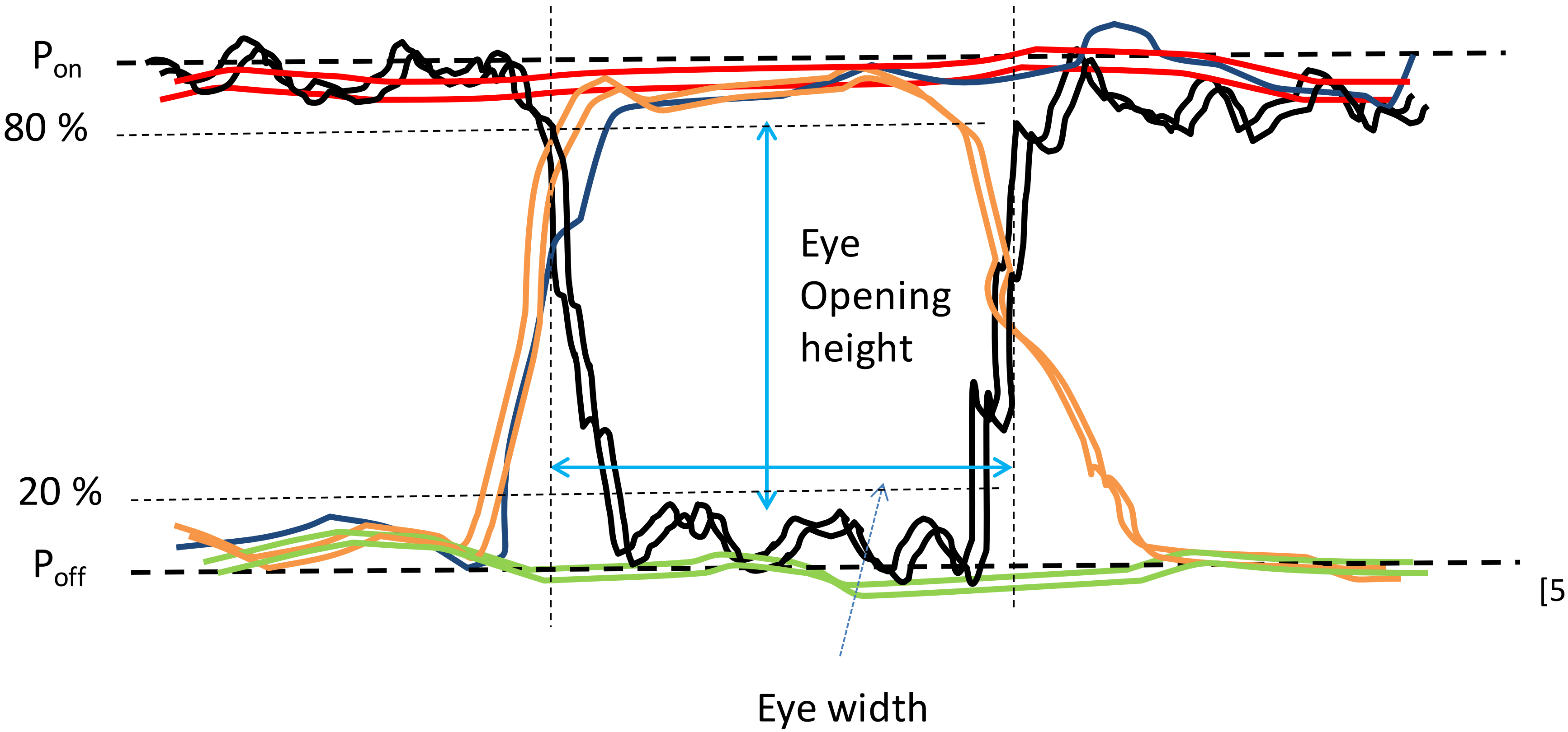


Eye diagram analysis: eye height & width

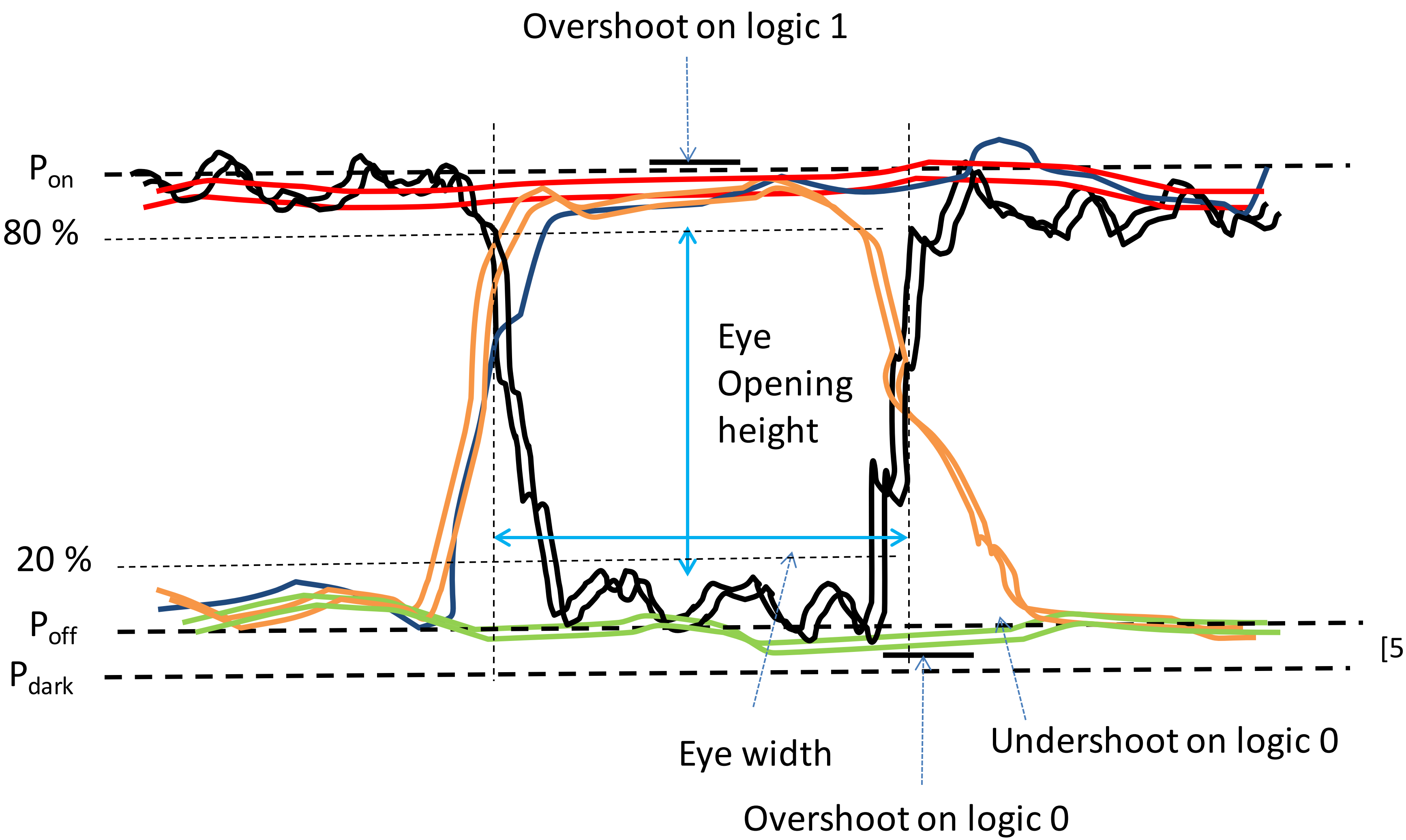


[5]

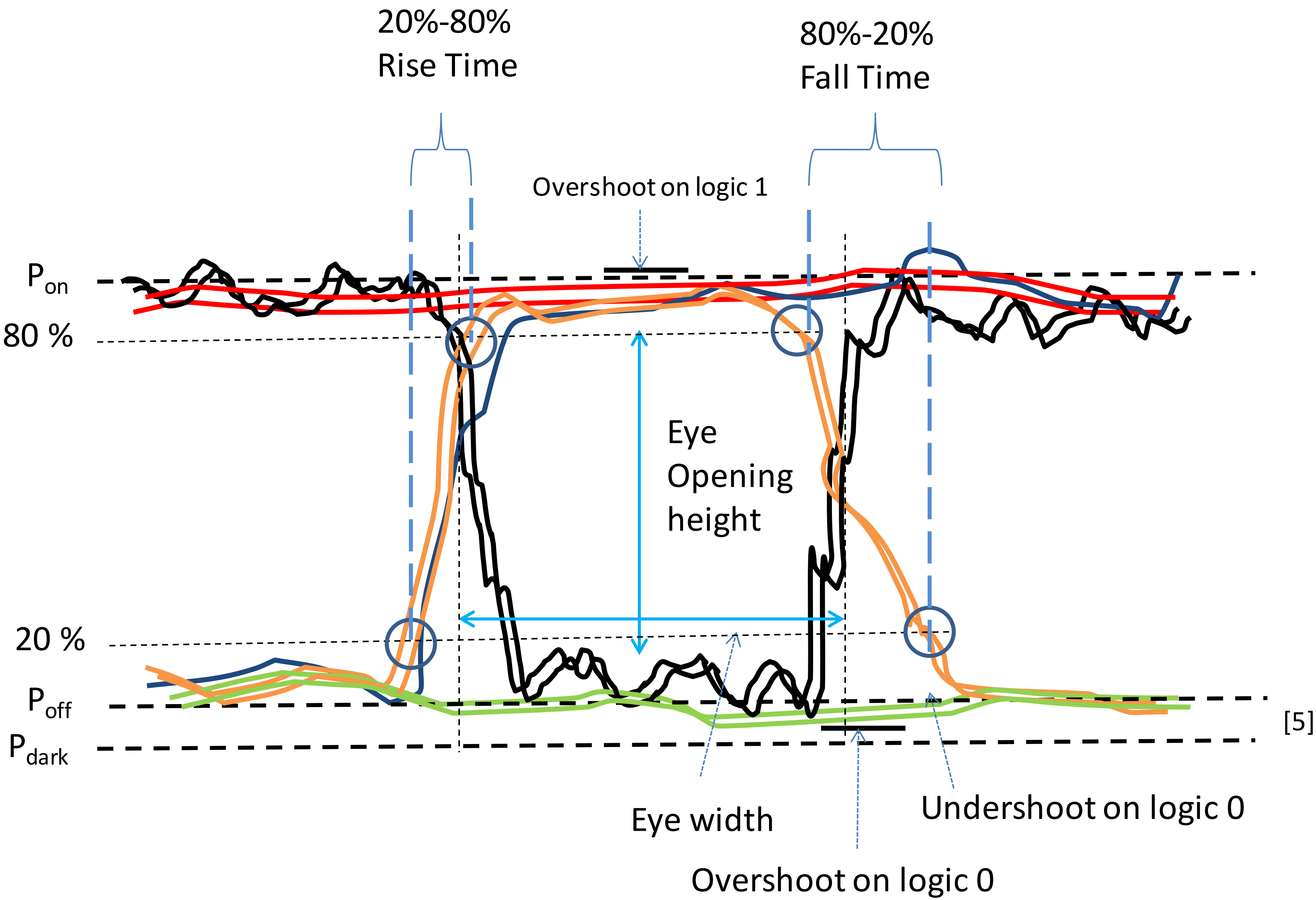
Eye diagram analysis: set the 20% & 80% traces



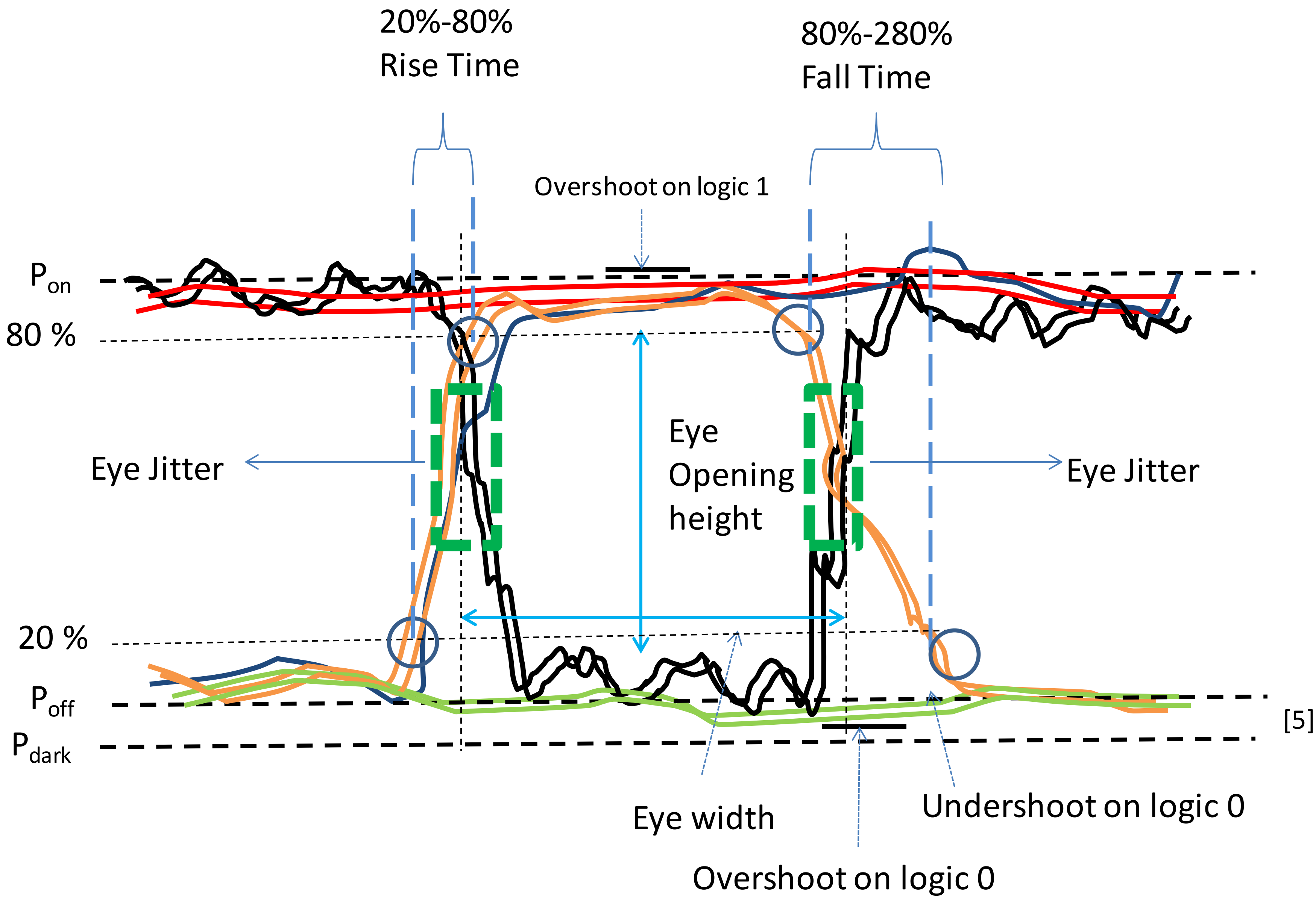
Eye diagram analysis: logic 1/0 levels



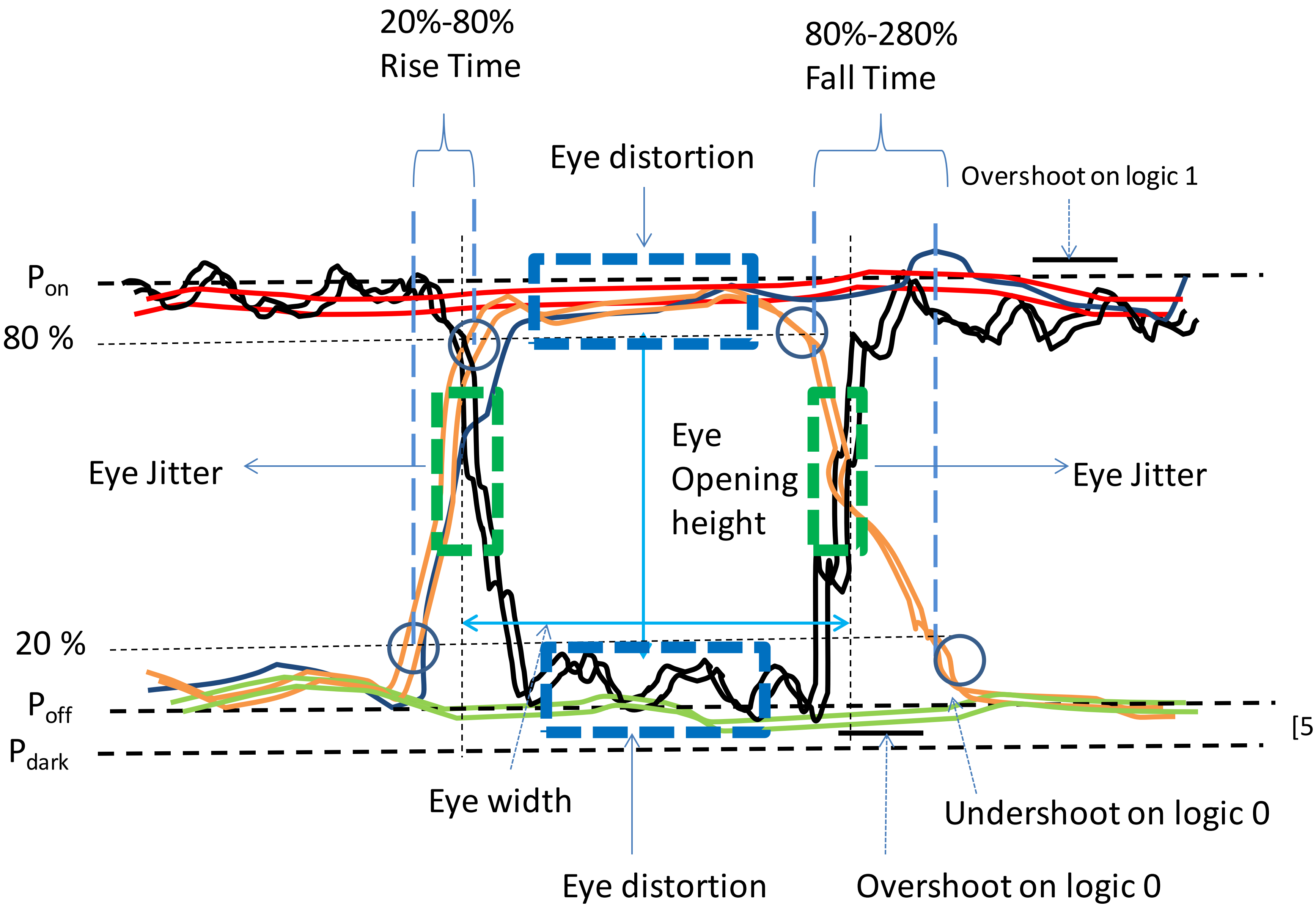
Eye diagram analysis: rise & fall time



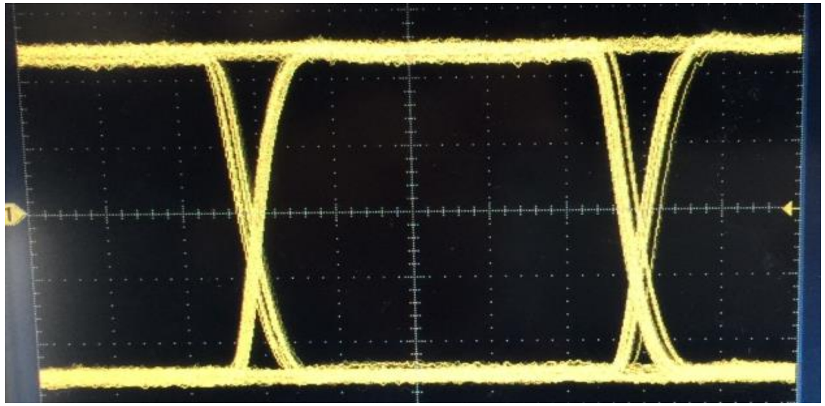
Eye diagram analysis: eye jitter



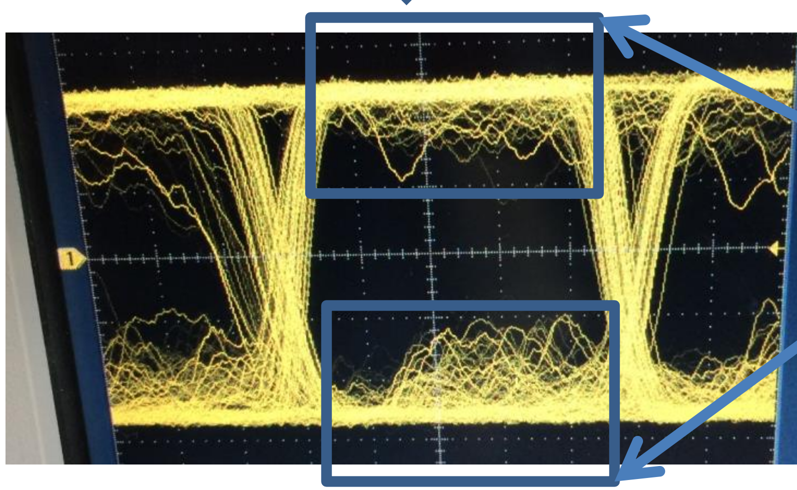
Eye diagram analysis: eye distortion



Distortion effects of an eye



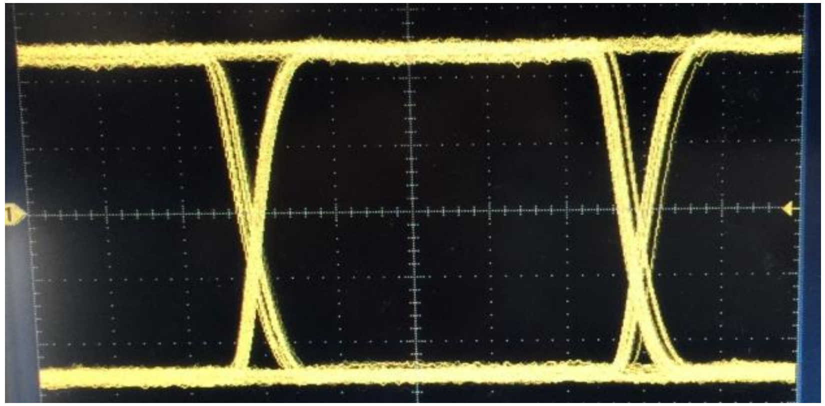
Makes Eye Diagram Opening Smaller [5]



BER ↑

Distorting effects

Jitter effects of an eye



Makes Eye Diagram Opening Smaller [5]

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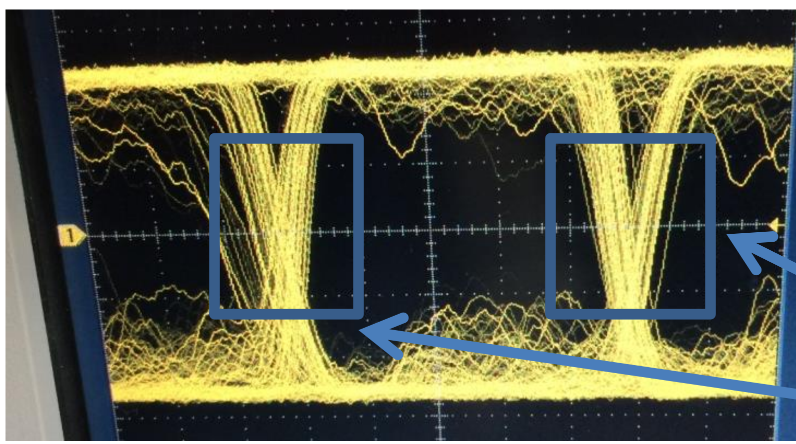
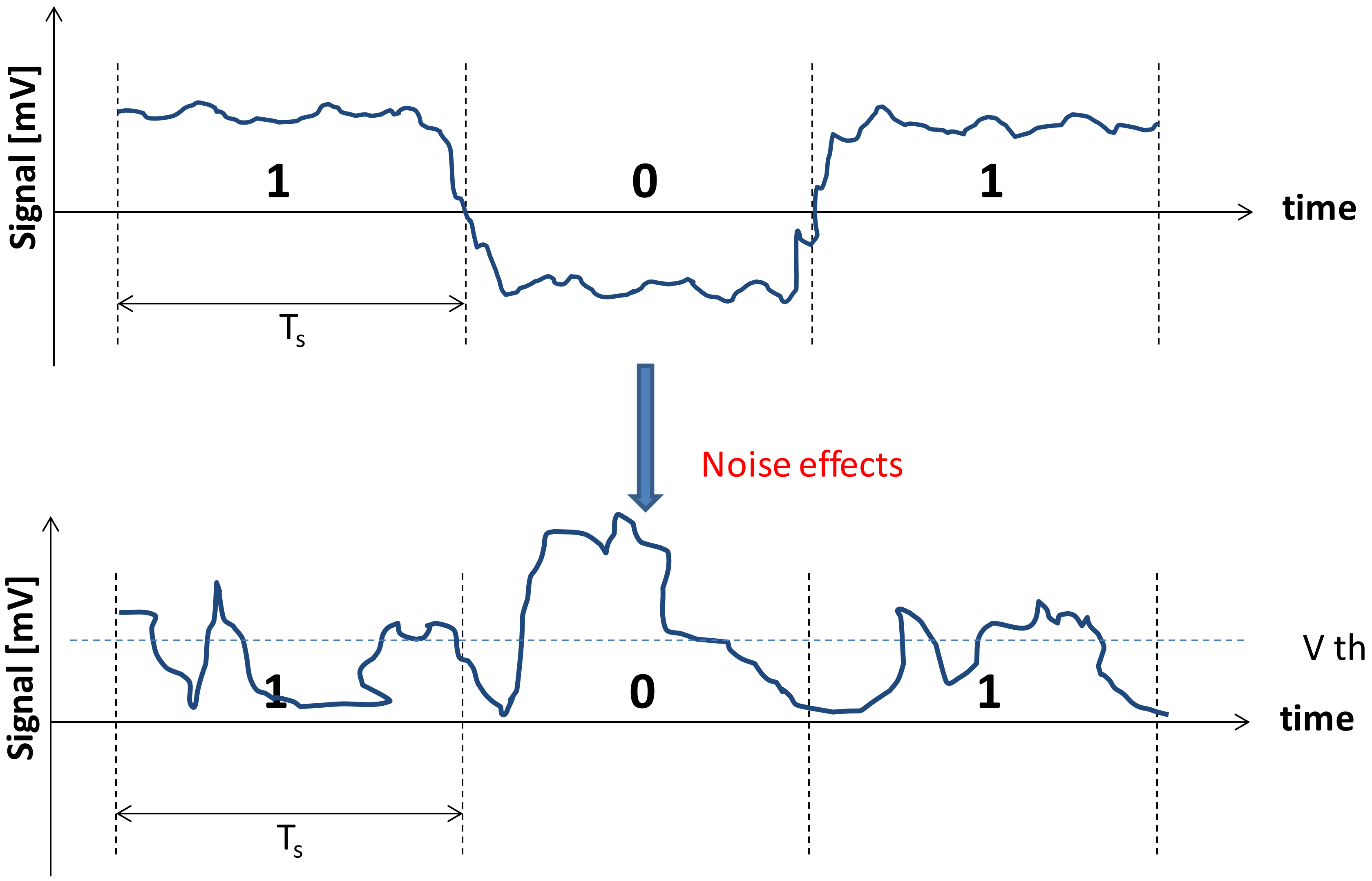


Jitter effects

Noise : basic concept



- Example: NRZ

Time Jitter: basic concept in optical systems

Timing jitter (edge jitter or phase distortion comes from Noise in the receiver and pulse distortion in optical systems [5]

Examples for Noise in a receiver

1- preamplifier noise at the receiver

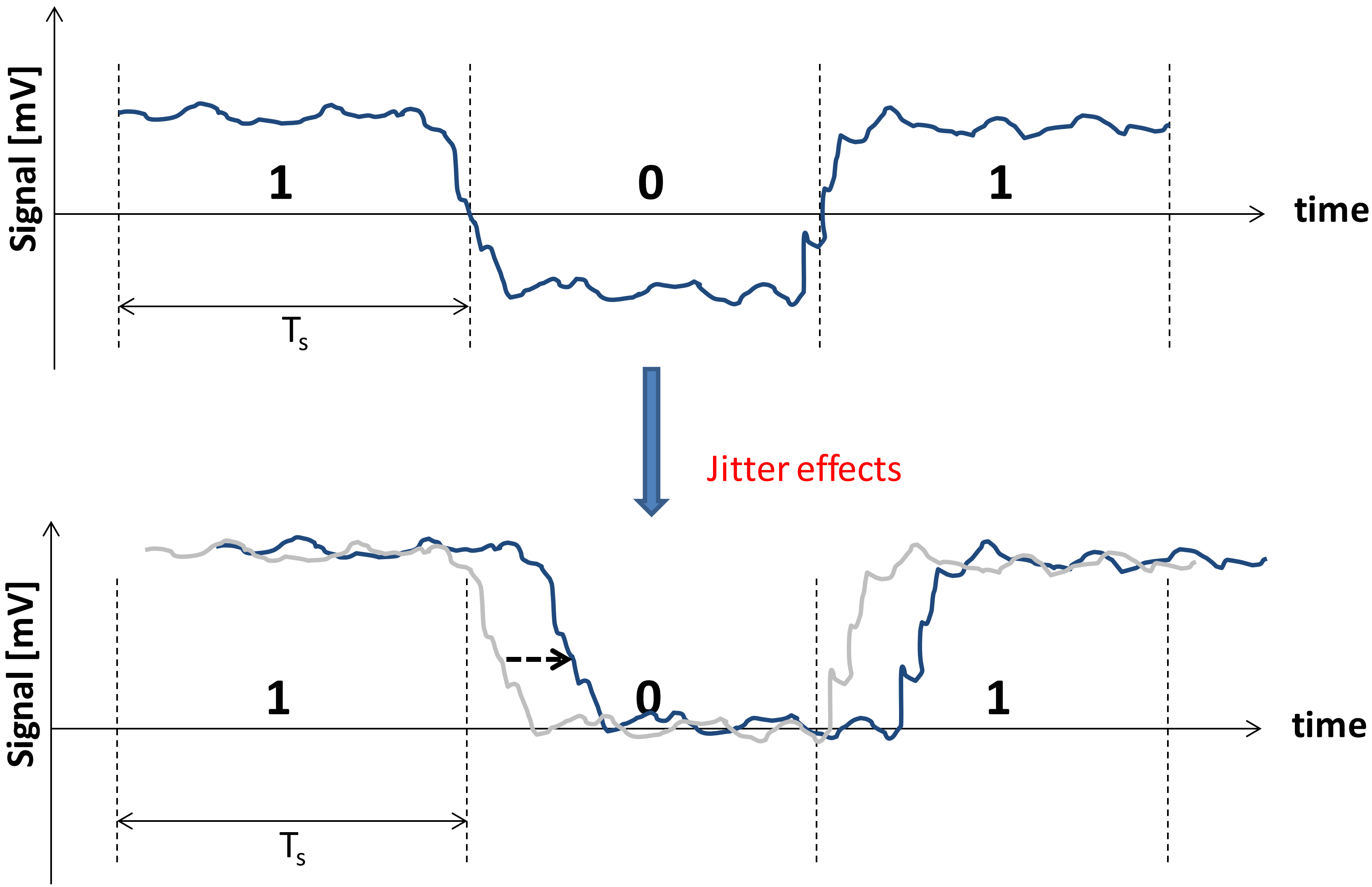
2- Photodectror noise (e.g. thermal noise)

Examples for Pulse distortion in optical systems

1- Material dispersion

2- Waveguide dispersion

Time Jitter: basic concept



Timing jitter (edge jitter or phase distortion comes from Noise

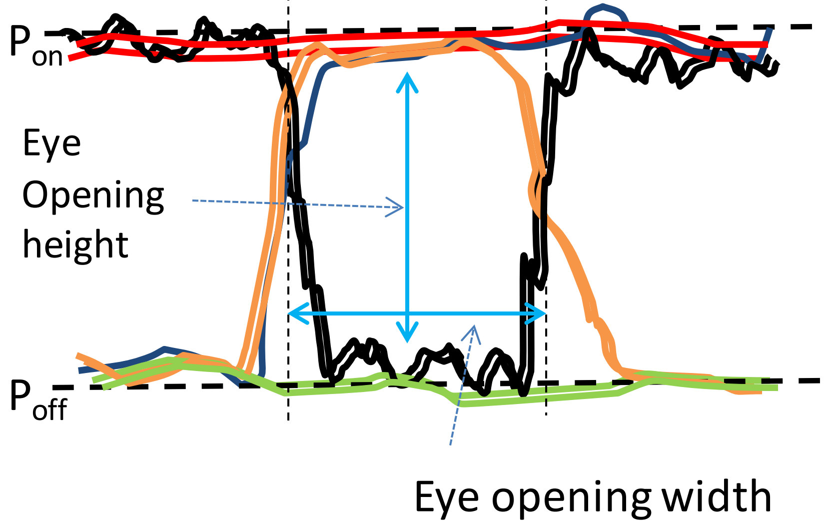
- Example: NRZ

What we care for eye diagram analysis

<u>We Care about</u>

- ✓ Jitter
- ✓ Noise
- ✓ Waveguide dispersion
- ✓ Material dispersion
- ✓ Loss

Impact Quality of optical system

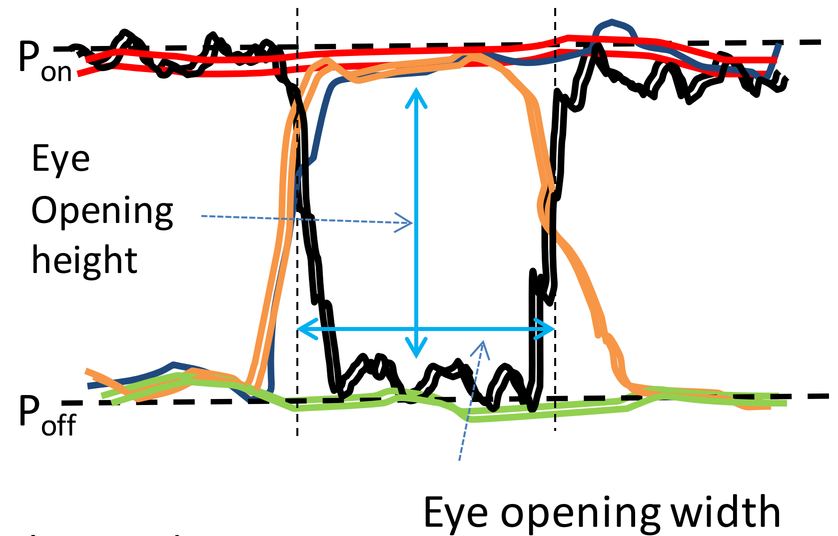


What we care for eye diagram analysis

We Care about

- ✓ Jitter
- ✓ Noise
- ✓ Waveguide dispersion
- ✓ Material dispersion
- ✓ Loss

Impact Quality of optical system



Where the impact of the above impairments can be observed

1- Qualitative observe

- ✓ Eye fall and rise time
- ✓ Crossing time
- ✓ Eye opening width

- ✓ Eye amplitude
- ✓ Crossing amplitude
- ✓ Eye opening height

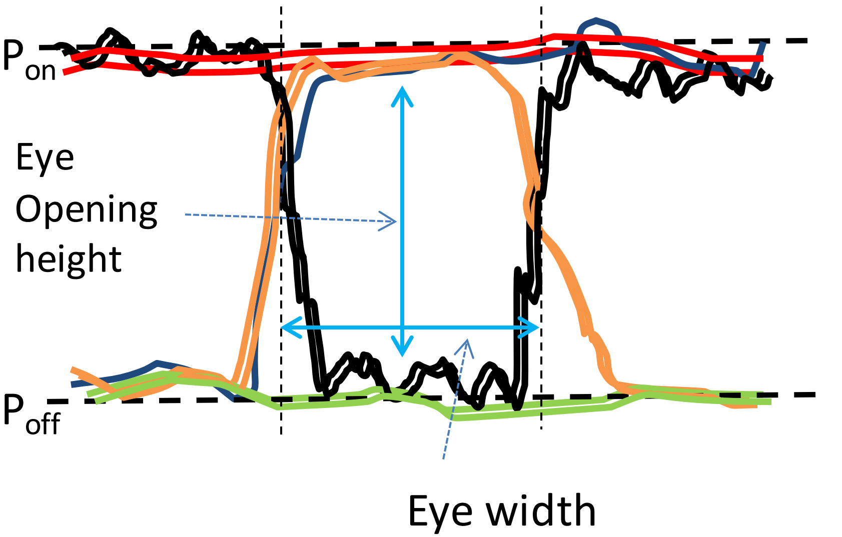
2- Quantitatively observe

- ✓ BER
- ✓ Q -factor

What we care for eye diagram analysis

How to Influence for Si waveguide

- ✓ Shape
- ✓ Geometry
- ✓ Dimensions
- ✓ Materials
- ✓ Index Contrast



What we care for eye diagram analysis

How to Influence for Si waveguide

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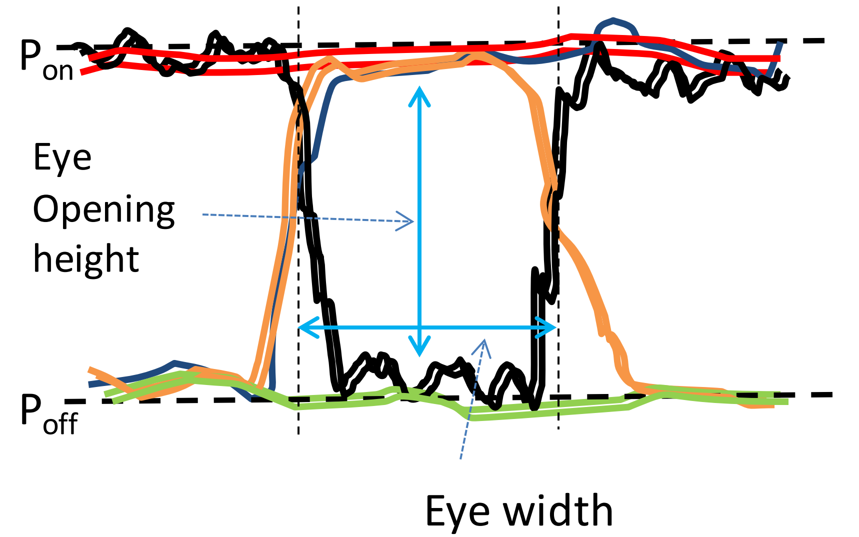
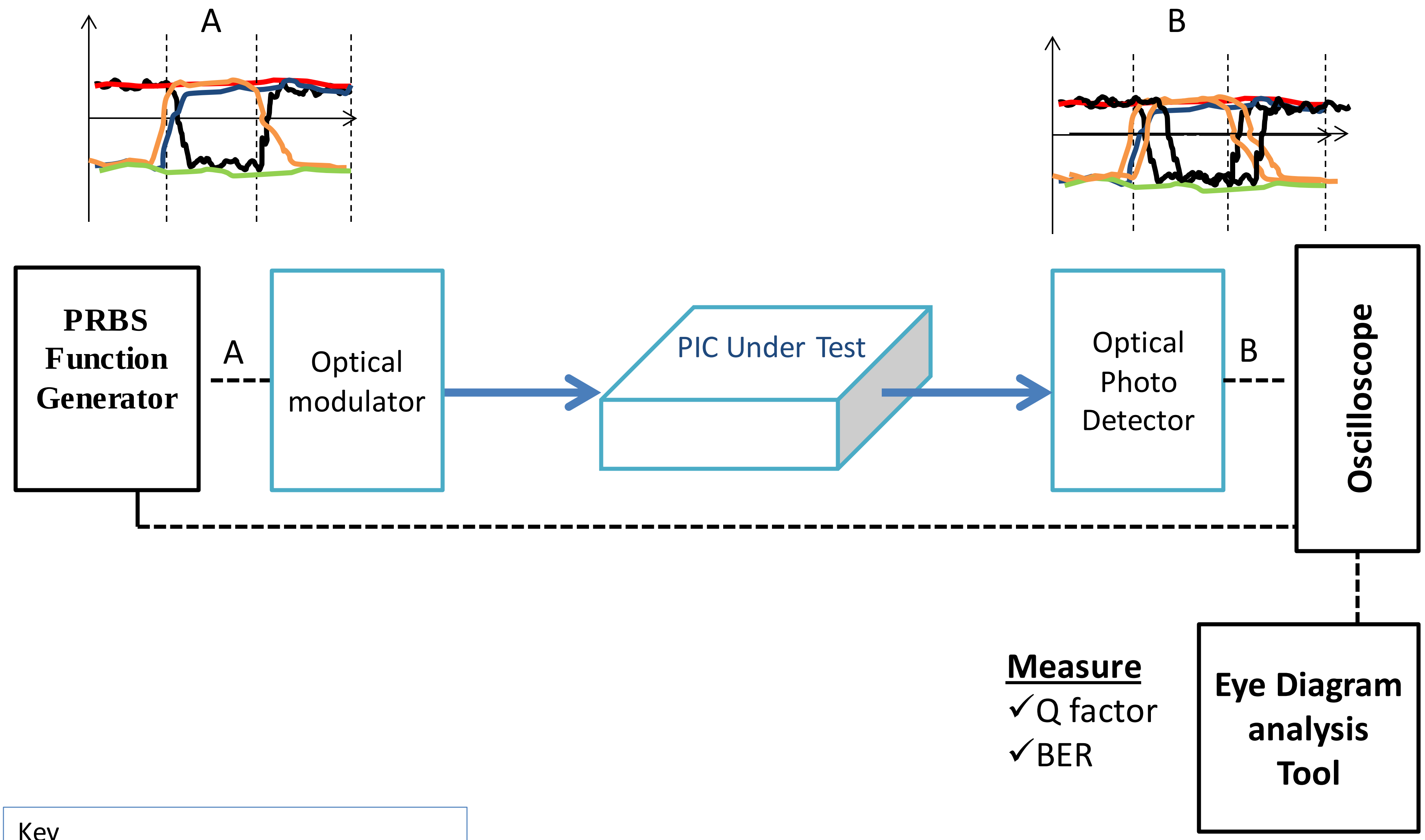


How to Influence for optical Receiver

- ✓ Receiver sensitivity
- ✓ Receiver noise
- ✓ Low preamplifier noise
- ✓ Materials
- ✓ shape
- ✓ Using DSP (e.g. FEC)

Schematic diagram of measuring eye diagram for a PIC

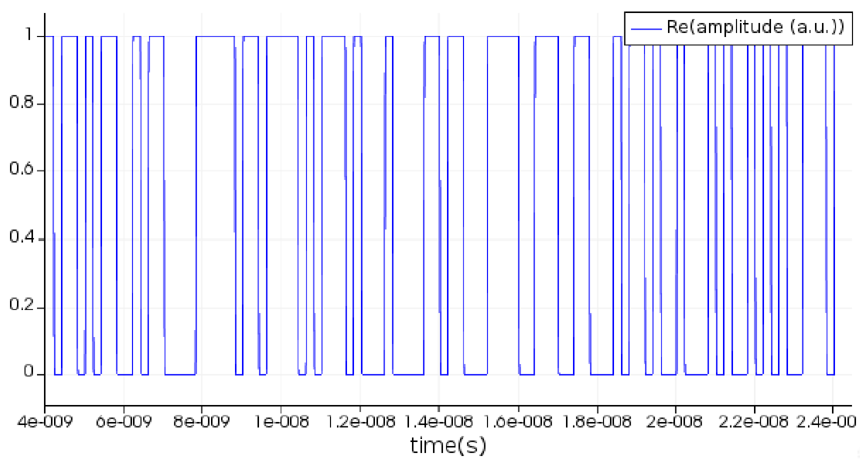


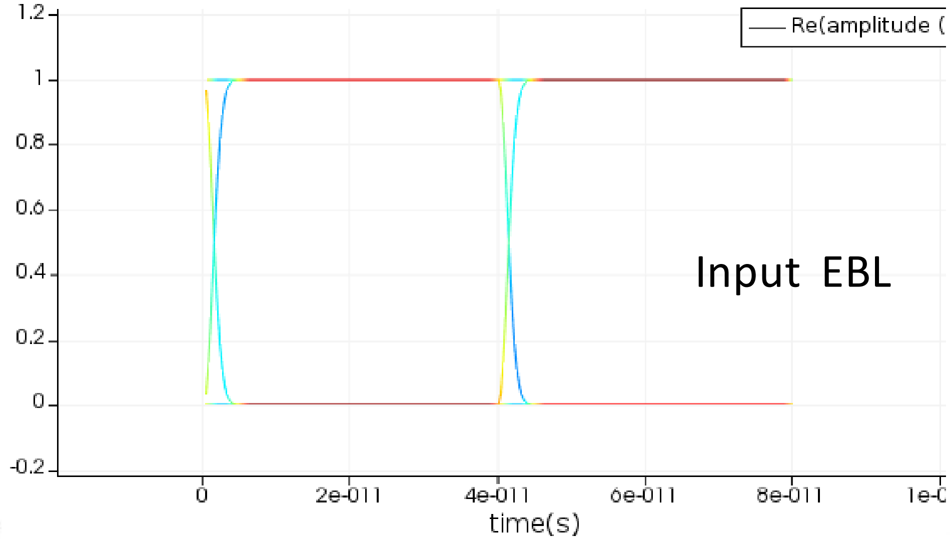
Measure
✓Q factor
✓BER

Key
Optical cable
Electrical cable

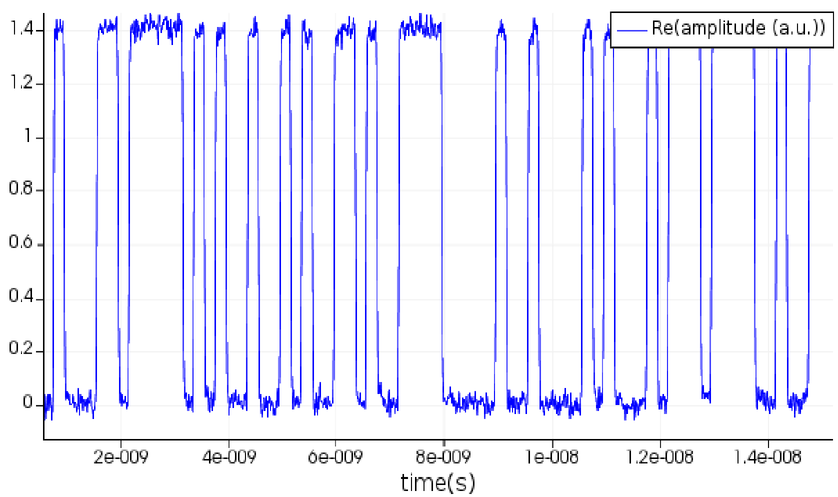
Eye diagram of NRZ Traveling Wave Modulators (5 Gb/s) using Ideal component Lumerical Interconnect [6]

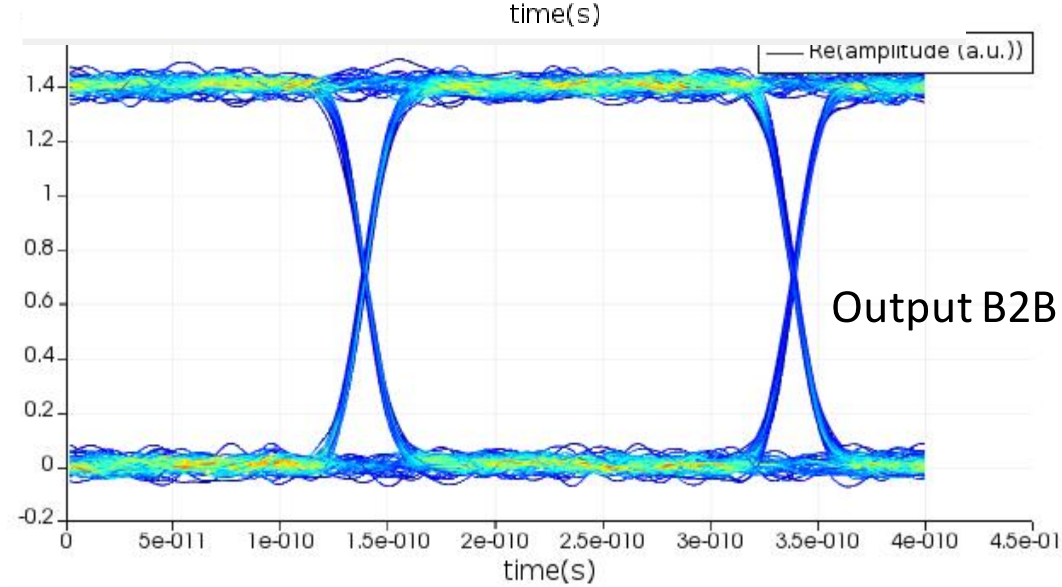
PRPS data at Input





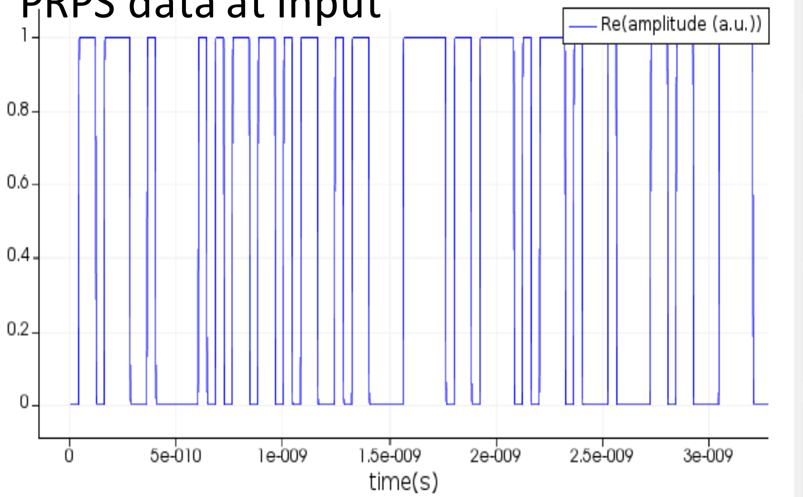
PRPS data at output

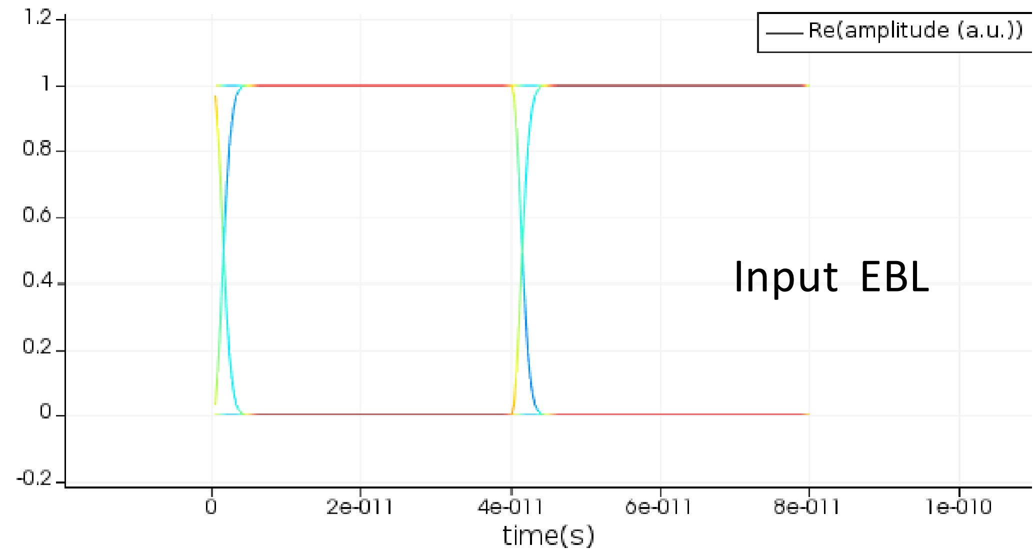




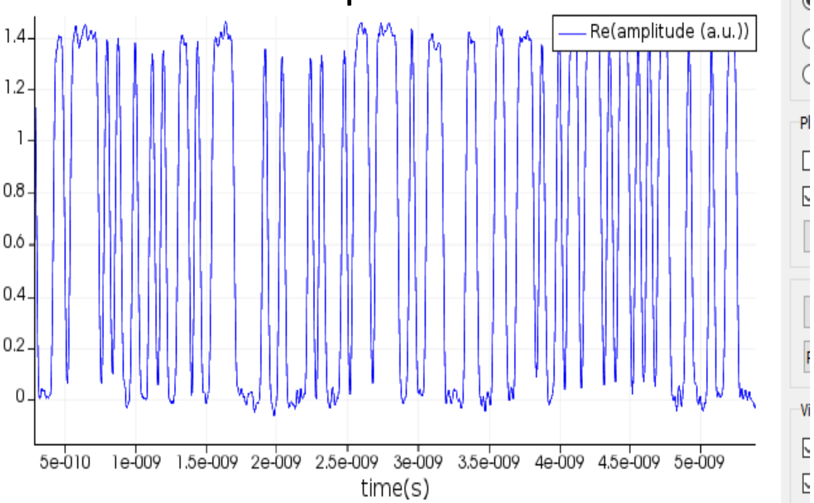
Eye diagram of NRZ Traveling Wave Modulators (25 Gb/s) using Ideal component Lumerical Interconnect [6]

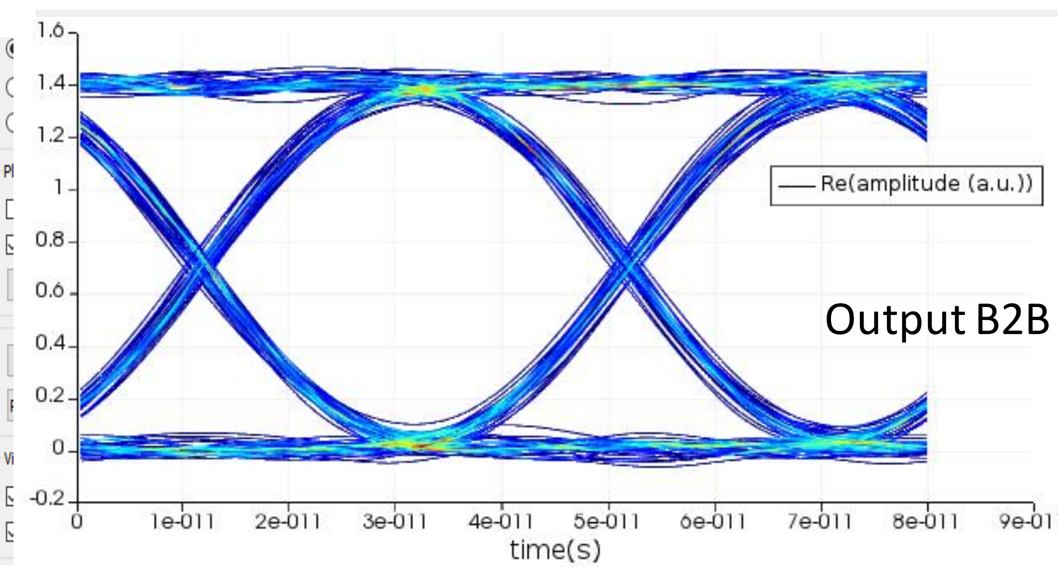
PRPS data at Input





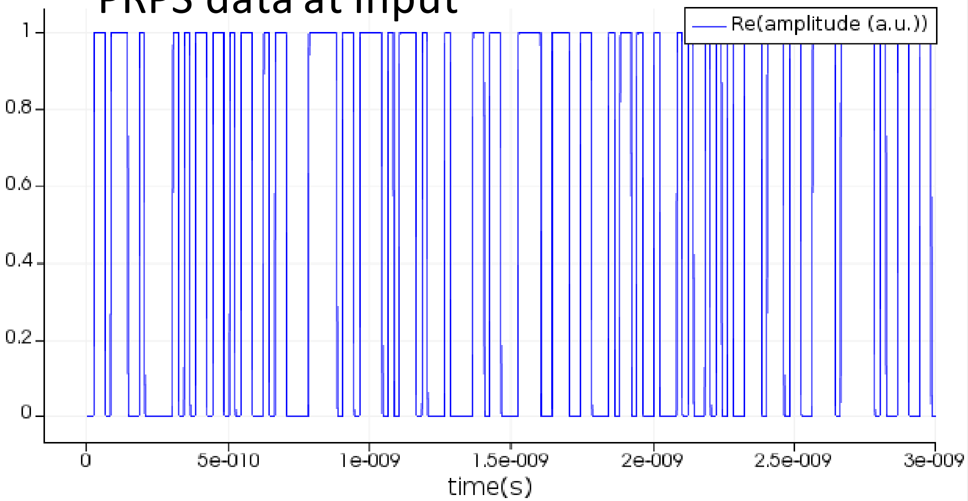
PRPS data at output

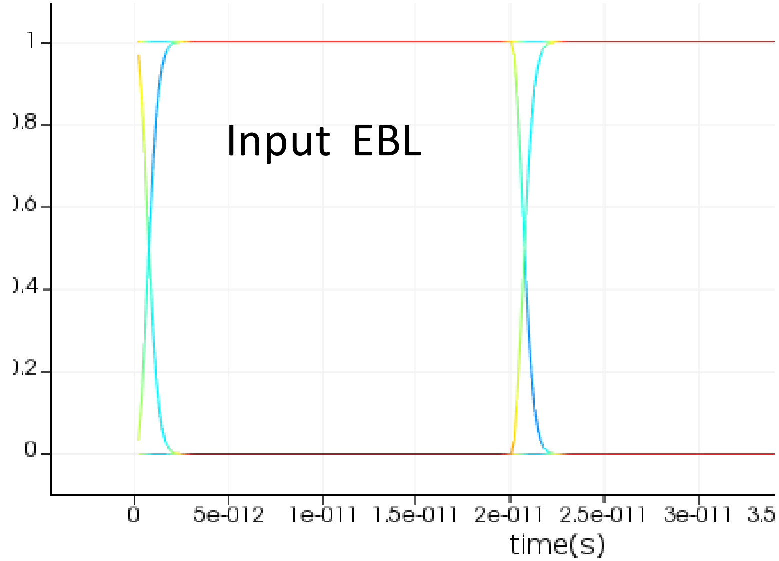




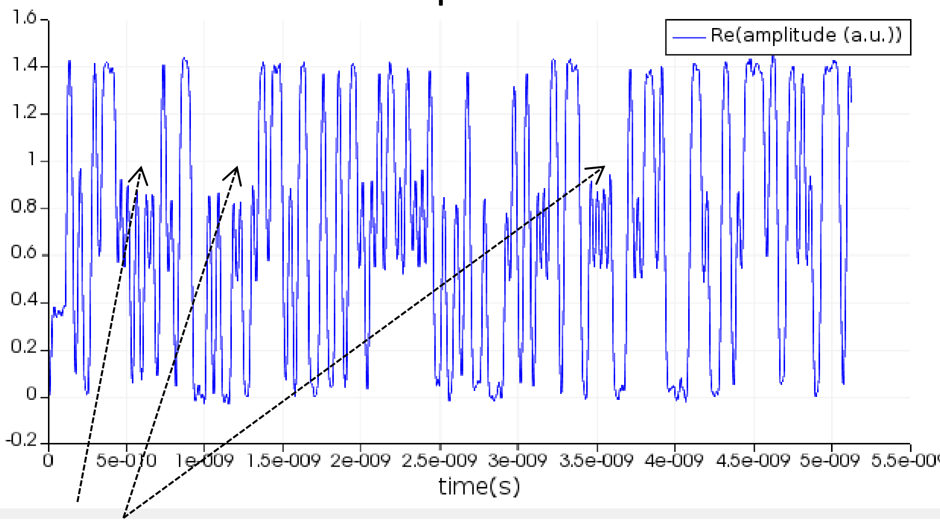
Eye diagram of NRZ Traveling Wave Modulators) (50 Gb/s) using Ideal component Lumerical Interconnect [6]

PRPS data at Input

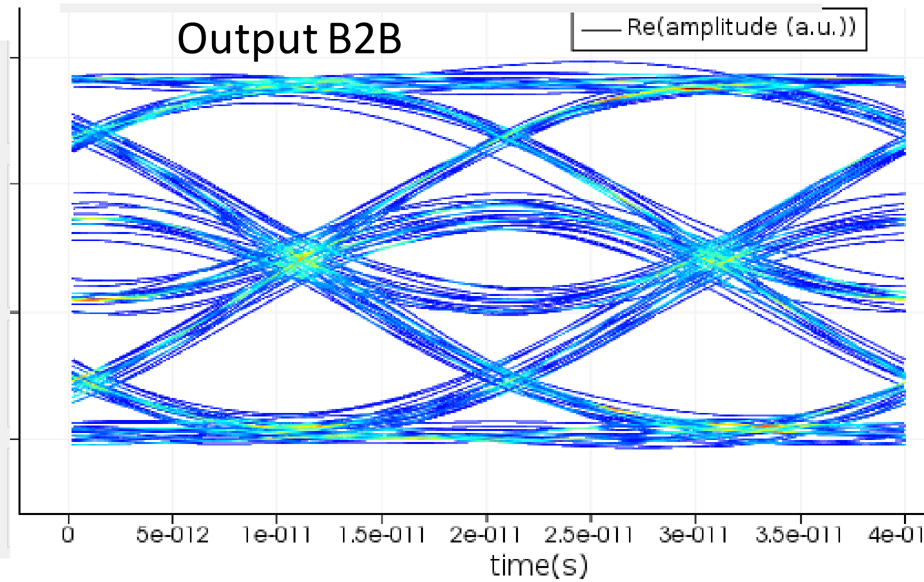




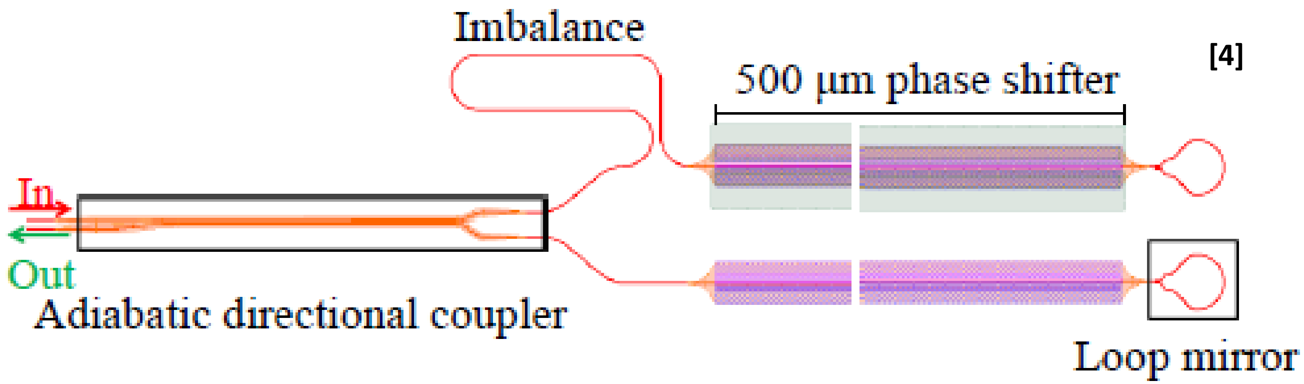
PRPS data at output



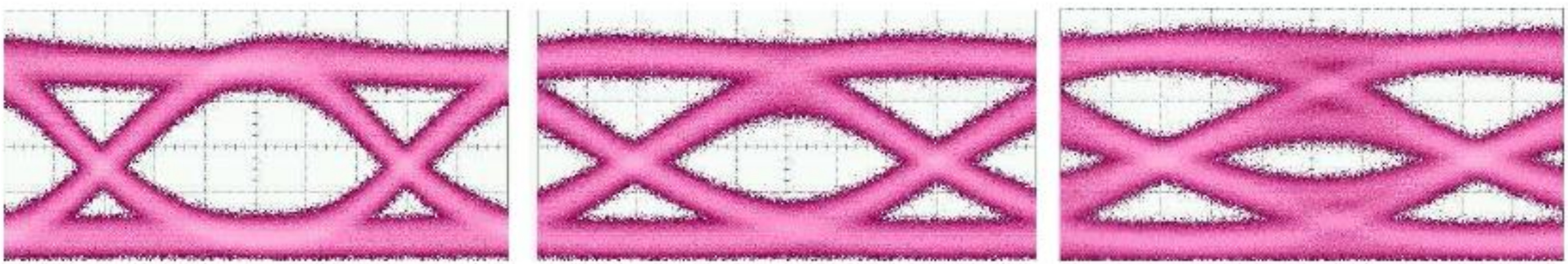
Errors



Eye diagrams for Photonic Device example "High-speed compact silicon photonic Michelson interferometric modulator [4]"



[4]



(a) ER: 6.6 dB, Q: 9.9
18 Gbps

(b) ER: 5.1 dB, Q: 5.5
26 Gbps

(c) ER: 4.0 dB, Q: 2.8
34 Gbps

[4]

BER & Q- Factor mathematical relation

$$BER = \frac{1}{2}\, erfc\ \frac{Q}{\sqrt{2}}$$ [3]

BER & Q- Factor mathematical relation

$$BER = \frac{1}{2}\, erfc\, \frac{Q}{\sqrt{2}}$$ [3]

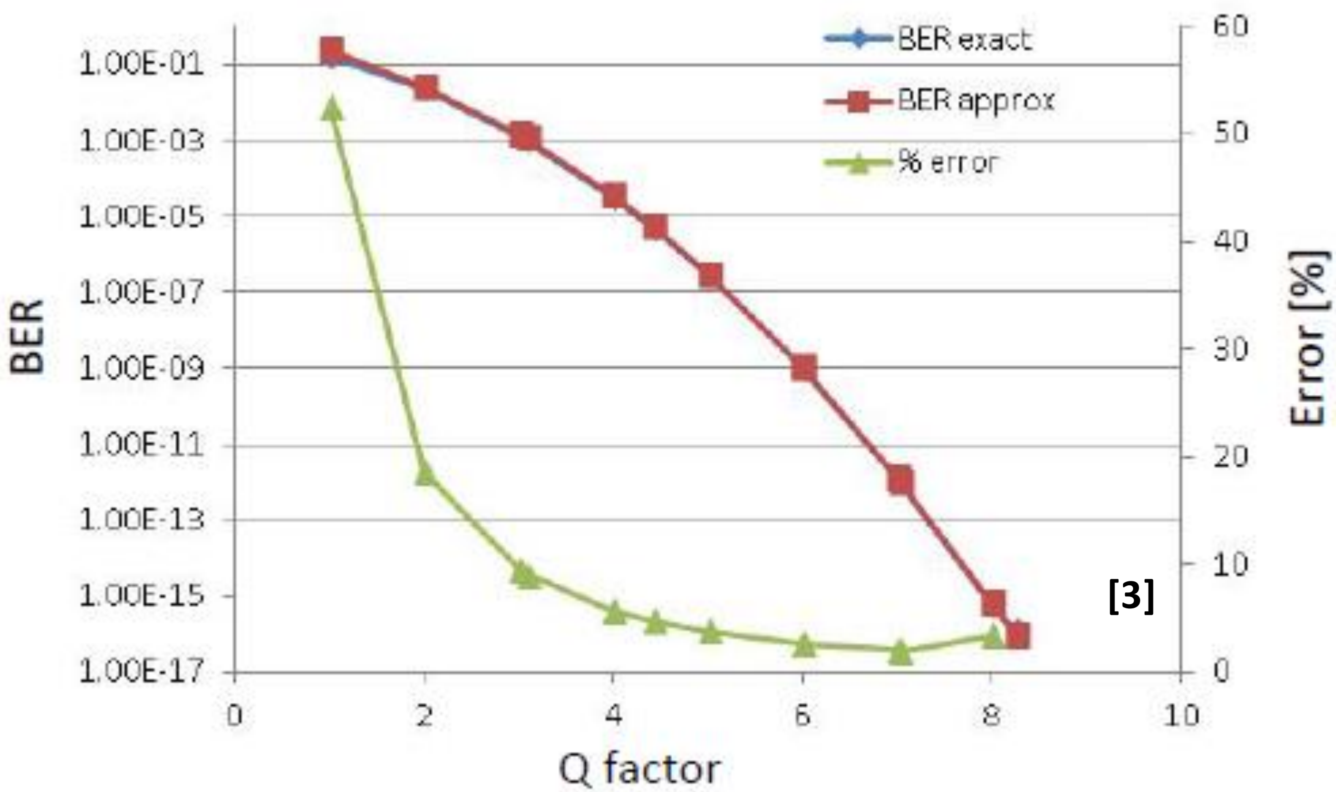
Q-Factor	BER
1	1.59E-01
2	2 2.28E-02
3	1.00E-03
3.09	1.35E-03
4	3.17E-05
4.417	5.00E-06
5	2.87E-07
5.997	1.00E-09

B & Q- Factor Mathematical Relation (Simplified for Q >3)

$$BER \approx \frac{1}{Q\sqrt{2\pi}}\, exp\,(-\frac{Q^2}{2})$$ **[3]**

BER & Q- Factor Mathematical Relation (Simplified for Q >3)

$$BER \approx \frac{1}{Q\sqrt{2\pi}}\, exp\,(-\frac{Q^2}{2})$$ [3]



[3]

Q- Factor with different representations

- Q [--], linear

[3]

- Q [dB] = $10\ Log_{10}(linear\ Q)$

- Q^2 [--], linear

- Q^2 [dB] = $10\ Log_{10}(linear\ Q^2) = 20\ Log_{10}(linear\ Q)$

References

[1] Chrostowski, Lukas, and Michael Hochberg. *Silicon Photonics Design: From Devices to Systems*. Cambridge University Press, 2015.

[2] Dr. Lukas Chrostowski Integrated Photonics II short course Silicon Photonics Design, Fabrication and Data Analysis. edX Online and summer course. UBCx: Phot1x Silicon Photonics Design, Fabrication and Data Analysis.

[3] Fiber Optic Telecommunications Technology TCET 740 \ TCET 745 Lecture- RIT \ *Fiber Optic Communications,* J. Downing, Delmar Learning, 2005

[4] Patel, David, Venkat Veerasubramanian, Samir Ghosh, Alireza Samani, Qiuhang Zhong, and David V. Plant. "High-speed compact silicon photonic Michelson interferometric modulator." *Optics express* 22, no. 22 (2014): 26788-26802.

[5] Keiser, Gerd. *Optical fiber communications*. John Wiley & Sons, Inc., 2003.

[6] Lumerical solutions
https://kb.lumerical.com/en/pic_circuits_traveling_wave_modulators.html